Group II, Row IV: extraction of the BFKL, JIMWLK and finite (DGLAP + RC) terms

1 Check of the starting expression

The starting expression is

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^++\Lambda}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,y,y',x'} e^{-ik\,(y'-y)\,\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)}\right]\frac{1}{\xi^2}$$
$$\left[\frac{1}{\epsilon}+\gamma_E+\frac{1}{2}\,\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\,\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)$$
$$+\;\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^++\Lambda}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,y,y',x'} e^{-ik\,(y'-y)\,\frac{1}{\xi}}\,\frac{1}{\xi^2}\left[\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\,\xi(1-\xi)\right.$$
$$\left.+\;\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\,\left[\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)}\right]\right]\,\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)\,. \qquad (1)$$

It is correct: it coincides exactly with the reduction of the Row-IV master expression (colour algebra $U^{d'b}(z)U^{db}(z)=\delta^{d'd}$, $f^{c'da}f^{cda}=N_c\delta^{c'c}$; δ-function integration over w,w'; contraction of the two tensor brackets over i,j, in which the antisymmetric $\frac{2}{k^+}(\delta_{mk'}\delta_{m'k}-\delta_{m'k'}\delta_{km})$ channel cancels; the dimensionally regularized z-integral $\int_z \mathcal{K}^m(y-z)\mathcal{K}^{m'}(y'-z)=-\frac{\pi}{2}L_\epsilon\delta^{mm'}-\pi\frac{r^m r^{m'}}{r^2}$, $r=y'-y$, $L_\epsilon\equiv\frac{2}{\epsilon}+2\gamma_E+\log\frac{r^2\mu^2_{\overline{MS}}}{4}$; and the substitution $\xi=\frac{k^+}{p^++k^+}$), as derived step by step in the companion note *"Group II, Row IV: reduction of the master expression"*. Two features were checked explicitly: (i) the ξ-limits are written *descending*, from $\xi_{\max}=\frac{k^+}{k^++\Lambda}$ down to $\bar\xi=\frac{k^+}{\vee}$, which absorbs the sign of $dp^+=-\frac{k^+d\xi}{\xi^2}$, so all the printed signs are right; (ii) the placement of $d_\perp$ is right: it multiplies the full $\mathcal{P}$-combination in the L_ϵ term, but only the azimuthal kernel in the second term — the dipole piece $\left[\frac{1-\xi}{\xi}+\frac{\xi}{1-\xi}\right]$ carries no $d_\perp$, since it originates from the $-\pi\,\delta_{mm'}\frac{r^m r^{m'}}{r^2}=-\pi$ trace of the z-integral.

2 Notation and first regrouping

Write $C\equiv\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}=\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}$, $f(\xi)=e^{-ik(y'-y)/\xi}$, $\hat p(\xi)=\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{1-\xi}$, and keep the colour bracket of (1) implicit. Flipping the limits to ascending ($\int_{\xi_{\max}}^{\bar\xi}=-\int_{\bar\xi}^{\xi_{\max}}$), Eq. (1) reads

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\,C\int_{\bar\xi}^{\xi_{\max}}d\xi\,\Big\{\frac{d_\perp}{2}\,\frac{\hat p(\xi)}{\xi^2}\,L_\epsilon+\frac{1}{\xi^2}\Big[\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big]\Big\}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,f(\xi)-C\,d_\perp\int_{\bar\xi}^{\xi_{\max}}d\xi\,\frac{1-\xi}{\xi}\,\frac{[\mathcal{K}(x-y)\cdot r][\mathcal{K}(x'-$$
(2)

Using the exact identity

$$\frac{\hat p(\xi)}{\xi^2}-\frac{1-\xi}{\xi}=\frac{1}{\xi^2}\Big[\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big]\,, \qquad (3)$$

the L_ϵ-free dipole piece is rewritten as $\hat p/\xi^2$ minus $\frac{1-\xi}{\xi}$, giving

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\,C\int_{\bar\xi}^{\xi_{\max}}d\xi\,\frac{\hat p(\xi)}{\xi^2}\,\underbrace{\Big[\frac{d_\perp}{2}L_\epsilon+1\Big]}_{=\,L_\epsilon\,+\,O(\epsilon)}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')f(\xi)\;+\;C\int_{\bar\xi}^{1}d\xi\,\frac{1-\xi}{\xi}\Big[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')-d_\perp\,\frac{(\mathcal{K}\cdot r)(\mathcal{K}'\cdot r)}{r^2}\Big]f(\xi)$$
(4)

where $\frac{d_\perp}{2}L_\epsilon+1=\frac{d_\perp}{\epsilon}+d_\perp\gamma_E+\frac{d_\perp}{2}\log\frac{r^2\mu^2}{4}+1=L_\epsilon$ for $d_\perp=2-\epsilon$ (the $+1$ from $-\frac{\epsilon}{2}\cdot\frac{2}{\epsilon}$ cancels the explicit $+1$), and the upper limit of the regular second integral has been extended to 1 at a cost $O(\Lambda)$.

3 Endpoint extraction: the JIMWLK term

The only remaining Λ-sensitivity of (4) sits in the $\frac{\xi}{1-\xi}$ channel of $\hat{p}$ at $\xi \to \xi_{\max}$ (i.e. $p^+ \to \Lambda$). With $h(\xi) \equiv \left[\frac{d_\perp}{2} L_\epsilon + 1\right] f(\xi)/\xi$ the endpoint lemma reads

$$\int_{\bar\xi}^{\xi_{\max}} d\xi \, \frac{h(\xi)}{1-\xi} = \int_{\bar\xi}^{1} d\xi \, \frac{h(\xi)-h(1)}{1-\xi} \; + \; h(1) \, \log \frac{(\vee - k^+)\, k^+}{\vee \Lambda} \; + \; O(\Lambda)\,, \qquad \log \frac{(\vee - k^+)k^+}{\vee \Lambda} = \log \frac{k^+}{\Lambda} + \log\left(1 - \frac{k^+}{\vee}\right), \tag{5}$$

with $h(1) = \left[\frac{d_\perp}{2} L_\epsilon + 1\right] e^{-ik(y'-y)}$. The $\log \frac{k^+}{\Lambda}$ part is the **JIMWLK contribution of this row**,

$$\boxed{\frac{d^3 N^{\text{JIMWLK}}_{\text{G-II, R IV}}}{d^3k} = +\frac{1}{(2\pi)^3} \frac{g^4 N_c}{8\pi^4} \frac{1}{k^+} \log\left(\frac{k^+}{\Lambda}\right) \int_{x,y,x',y'} e^{-ik\cdot(y'-y)} \, \mathcal{K}(x-y)\cdot\mathcal{K}(x'-y') \left[-\frac{d_\perp}{\epsilon} - d_\perp \gamma_E - \frac{d_\perp}{2} \log \frac{(y'-y)^2 \mu^2_{\overline{MS}}}{4} - 1 \right.}$$

(6)

(the bracket is $-L_\epsilon$), which is moved to the two-ρ evolution (JIMWLK) file, where it appears verbatim. The $\log(1-\frac{k^+}{\vee})$ part of (5) stays here.

4 The standard plus prescription: DGLAP and RC terms

With the *standard* plus distribution on $[0,1]$,

$$\int_a^1 d\xi \, \frac{h(\xi)}{(1-\xi)_+} = \int_a^1 d\xi \, \frac{h(\xi)-h(1)}{1-\xi} + h(1) \, \log(1-a)\,, \qquad a = \bar\xi = \frac{k^+}{\vee}\,, \tag{7}$$

the locally subtracted integral of (5) and the leftover $h(1)\log(1-\frac{k^+}{\vee})$ term recombine into the single integral $\int_{\bar\xi}^1 d\xi \, h(\xi)/(1-\xi)_+$: the two groupings differ only by where the finite piece $h(1)\log(1-\bar\xi)$ is placed. This conversion changes nothing, but it is required for

$$\mathcal{P}_{gg,+}(\xi) = 2N_c \left[\xi(1-\xi) + \frac{1-\xi}{\xi} + \frac{\xi}{(1-\xi)_+}\right], \qquad \mathcal{P}_{gg}(\xi) = \mathcal{P}_{gg,+}(\xi) + \frac{\beta_0}{2}\,\delta(1-\xi)\,, \qquad \beta_0 = \frac{11 N_c}{3} \text{ (pure glue)} \tag{8}$$

to be the *standard* splitting function. Using $C = \frac{g^2}{8\pi^2} \cdot 2N_c \cdot \frac{1}{(2\pi)^3} \frac{g^2}{2\pi^2} \frac{1}{k^+}$ and defining

$$\frac{d^3 N_{\xi\,\text{LO}}}{d^3k} = \frac{1}{(2\pi)^3} \frac{g^2}{2\pi^2} \frac{1}{k^+} \int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}} \, \mathcal{K}(x-y)\cdot\mathcal{K}(x'-y') \, \left[U^{ce'}(y') - U^{ce'}(x')\right] \left[U^{ce}(y) - U^{ce}(x)\right] \rho^{e'}(x')\, \rho^{e}(x)\,, \tag{9}$$

the first term of (4) minus the JIMWLK term becomes $-\frac{g^2}{8\pi^2} \int_{\bar\xi}^1 \frac{d\xi}{\xi^2} \, \mathcal{P}_{gg,+}(\xi)\, L_\epsilon \, \frac{d^3 N_{\xi\,\text{LO}}}{d^3k}$. Finally, adding and subtracting the $\frac{\beta_0}{2}\delta(1-\xi)$ piece,

$$-\frac{g^2}{8\pi^2} \int_{\bar\xi}^1 \frac{d\xi}{\xi^2} \, \mathcal{P}_{gg,+}\, L_\epsilon \, \frac{d^3 N_{\xi\,\text{LO}}}{d^3k} = \underbrace{-\frac{g^2}{8\pi^2} \int_{\bar\xi}^1 \frac{d\xi}{\xi^2} \, \mathcal{P}_{gg}(\xi)\, L_\epsilon \, \frac{d^3 N_{\xi\,\text{LO}}}{d^3k}}_{\text{DGLAP}} + \underbrace{\frac{g^2 \beta_0}{16\pi^2} \, L_\epsilon \, \frac{d^3 N_{\text{LO}}}{d^3k}}_{\text{RC}}\,, \tag{10}$$

which makes the running-coupling (RC) term explicit; the two added pieces cancel identically, so this is an exact add-and-subtract.

5 The soft subtraction: the BFKL term

The $2N_c \frac{1-\xi}{\xi}$ part of $\mathcal{P}_{gg}$ in the DGLAP term produces a hidden $\log \frac{\vee}{k^+}$ from $\xi \to 0$ (in the original variables this is the collapse $y = \xi w + (1-\xi) z \to z$, $y' = \xi w' + (1-\xi) z \to z$ of the parent positions onto the sibling at fixed w, w'), and a rapidity logarithm with an L_ϵ coefficient would not be physical. Its $\xi \to 0$ limit is subtracted *exactly* with

$$\frac{d^3 N^{\text{sub}}_{\xi}}{d^3k} = \frac{1}{(2\pi)^3} \frac{g^2}{2\pi^2} \frac{1}{k^+} \int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}} \, \frac{1}{\pi} \, K(y,y',z)\, K(x,x',z) \left[U^{ce'}(z) - U^{ce'}(x')\right] \left[U^{ce}(z) - U^{ce}(x)\right] \rho^{e'}(x')\, \rho^{e}(x)\,, \tag{11}$$

$K(a,b,z)=\frac{(a-z)\cdot(b-z)}{(a-z)^2(b-z)^2}$, which is IR and UV finite in $d=2$, contains no μ, and is symmetric under $(x,y)\leftrightarrow(x',y')$. Writing $W(u,u')=[U^{ce'}(u')-U^{ce'}(x')][U^{ce}(u)-U^{ce}(x)]\rho^{e'}(x')\rho^{e}(x)$ and using $\int_z^{(d=2-\epsilon)}K(y,y',z)=-\pi L_\epsilon$, the per-$\xi$ combination

$$2N_c\frac{1-\xi}{\xi}\Big[L_\epsilon\,\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}+\frac{d^3N_\xi^{\mathrm{sub}}}{d^3k}\Big]=-\frac{2N_c(1-\xi)}{\xi}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int e^{-ik(y'-y)\frac{1}{\xi}}\,\frac{K(y,y',z)}{\pi}\Big[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,W(y,y')-K(x,z$$

(12)

contains no $\frac{2}{\epsilon}$ and no μ; at fixed w,w' the square bracket is $O(\xi)$ as $\xi\to0$, so no $\log\vee$ survives with an L_ϵ coefficient. The exact value of the subtraction follows from $y=z+\xi(w-z)$, $y'=z+\xi(w'-z)$ (Jacobian ξ^4, $K(y,y',z)=K(w,w',z)/\xi^2$, phase $\to e^{-ik(w'-w)}$) and $\int_{\bar\xi}^1 d\xi\frac{1-\xi}{\xi}=\log\frac{\vee}{k^+}-(1-\frac{k^+}{\vee})$:

$$-\frac{g^2}{8\pi^2}\int_{\bar\xi}^1\frac{d\xi}{\xi^2}\,2N_c\,\frac{1-\xi}{\xi}\,\frac{d^3N_\xi^{\mathrm{sub}}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^5}\,\frac{1}{k^+}\Big[\log\frac{\vee}{k^+}-\Big(1-\frac{k^+}{\vee}\Big)\Big]\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}K(w,w',z)\,K(x,x',z)\,W(z,z)\,.$$

(13)

Adding and subtracting (13), the $\log\frac{\vee}{k^+}$ part is the **BFKL contribution of this row**,

$$\boxed{\frac{d^3N^{\mathrm{BFKL}}_{\text{G-II, R IV}}}{d^3k}=+\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)}$$

(14)

moved to the BFKL file (it appears there verbatim); its coefficient is manifestly UV finite and μ-independent, as a BFKL kernel must be. The $-\big(1-\frac{k^+}{\vee}\big)$ part remains as the boundary term below.

6 Result: the finite terms

Collecting everything, Eq. (1) decomposes as

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k}=\frac{d^3N^{\mathrm{BFKL}}}{d^3k}\Big|_{(14)}+\frac{d^3N^{\mathrm{JIMWLK}}}{d^3k}\Big|_{(6)}+\frac{d^3N^{\mathrm{finite}}}{d^3k}\,,\tag{15}$$

with the finite part containing the DGLAP and RC terms:

$$\begin{aligned}\frac{d^3N^{\mathrm{finite}}}{d^3k}=&\underbrace{-\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{g^2}{8\pi^2}\,\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,L_\epsilon\,\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-}_{\text{DGLAP}}\\&\underbrace{-\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y',z}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,K(y,y',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x}_{\text{soft subtraction}}\\&\underbrace{+\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{g^2\,\beta_0}{16\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,L_\epsilon\,\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)}_{\text{RC}}\\&\underbrace{-\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\Big(1-\frac{k^+}{\vee}\Big)\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x'}_{\text{boundary}}\\&\underbrace{+\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^4}\,\frac{1}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\right]}_{\text{azimuthal}}\end{aligned}\tag{16}$$

with $\mathcal{P}_{gg}$ in the *standard* $[0,1]$ plus-prescription convention (7)–(8), $L_\epsilon=\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}$, and $d_\perp\to2$ set in the azimuthal term (no pole multiplies it).

Bookkeeping

(i) (soft subtraction) + (boundary) + (BFKL entry (14)) = 0 identically — the construction is an exact add-and-subtract of (13); all three objects are UV finite. The physical origin of the BFKL logarithm is the $\xi\to0$ endpoint of the DGLAP term.

(ii) The $\frac{\beta_0}{2}\delta(1-\xi)$ piece inside $\mathcal{P}_{gg}$ cancels the RC term identically; the soft $2N_c\frac{1-\xi}{\xi}$ piece of the DGLAP term combines with the subtraction into the μ-free form (12). The only surviving ultraviolet logarithms are $L_\epsilon \otimes 2N_c\big[\frac{\xi}{(1-\xi)_+}+\xi(1-\xi)\big]$ — the physical collinear (DGLAP) kernel of the measured gluon, including its $(1-\xi)_+$ endpoint $\log(1-\frac{k^+}{\vee})$ piece — and the RC term; these enter the final pole accounting (coupling renormalization with $\beta_0=\frac{11N_c}{3}$ together with the Group-I ultraviolet terms, plus the collinear subtraction for the produced gluon).

(iii) Together with (6) and (14), Eq. (16) reproduces the starting expression (1) identically, up to $O(\Lambda)$ (endpoint recombination and extension of regular ξ-integrals to $\xi=1$) and $O(\epsilon)\times$ finite.